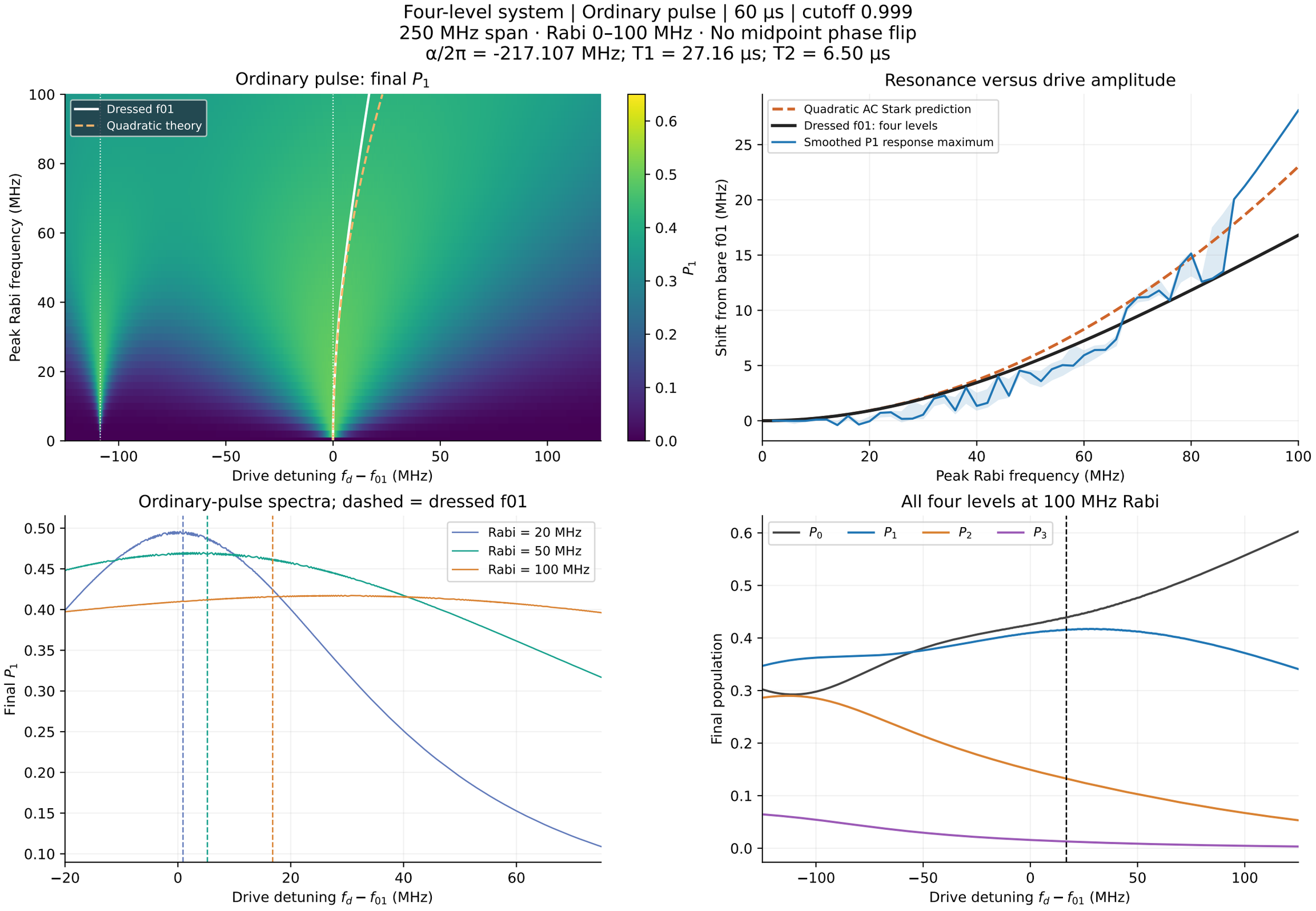

Four-level system | Ordinary pulse | 60 μs | cutoff 0.999
250 MHz span · Rabi 0–100 MHz · No midpoint phase flip
α/2π = -217.107 MHz; T1 = 27.16 μs; T2 = 6.50 μs
Ordinary pulse: final P_1
Dressed f01
Quadratic theory
Peak Rabi frequency (MHz)
Drive detuning $f_d - f_{01}$ (MHz)
P_1
Resonance versus drive amplitude
Quadratic AC Stark prediction
Dressed f01: four levels
Smoothed P1 response maximum
Shift from bare f01 (MHz)
Peak Rabi frequency (MHz)
Ordinary-pulse spectra; dashed = dressed f01
Rabi = 20 MHz
Rabi = 50 MHz
Rabi = 100 MHz
Final P_1
Drive detuning $f_d - f_{01}$ (MHz)
All four levels at 100 MHz Rabi
P_0
P_1
P_2
P_3
Final population
Drive detuning $f_d - f_{01}$ (MHz)